

C++ Fresher Viva Assessment Report

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Final Score: 80 / 100

Interview Performance: 24 / 30 questions

Q1: What is a constructor in C++?

Your Answer: A constructor is a special member function of a class that is automatically called when an object is created to initialize object.

Score Awarded: 2/2

Q2: What are the characteristics of a constructor?

Your Answer: No return type, has same name as class, modularity.

Score Awarded: 1.5/2

Q3: What is a destructor?

Your Answer: Deallocation of memory, destroys the class of the program.

Score Awarded: 1/2

Q4: Two differences between constructor and destructor.

Your Answer: Constructor: allocates memory, has same name as class. Destructor: deallocates memory, destroy the objects.

Score Awarded: 1.5/2

Q5: What is a default constructor?

Your Answer: Constructor without any parameters.

Score Awarded: 2/2

Q6: What is a parameterized constructor?

Your Answer: Constructor which consists of parameters.

Score Awarded: 2/2

Q7: What is a copy constructor?

Your Answer: Copies an already existing object into a new one.

Score Awarded: 2/2

Q8: What is operator overloading?

Your Answer: Objects can be added using overloaded operators such as +.

Score Awarded: 2/2

Q9: Which operators cannot be overloaded?

Your Answer: ., .*, :

Score Awarded: 1/2

Q10: Why is a destructor declared as ~ClassName()?

Your Answer: The ~ operator is used to delete the class.

Score Awarded: 0.5/2

Q11: Explain constructor overloading.

Your Answer: Different parameters or different data types.

Score Awarded: 2/3

Q12: Differentiate default, parameterized and copy constructors.

Your Answer: Default has no parameters, parameterized has parameters and copy copies an object.

Score Awarded: 3/3

Q13: Order of constructor and destructor execution.

Your Answer: Constructor then ~ operator destroys objects.

Score Awarded: 1/3

Q14: What happens if a copy constructor is not defined?

Your Answer: It will not copy the existing object into a new one.

Score Awarded: 2/3

Q15: Explain dynamic object creation.

Your Answer: new keyword is used to allocate memory dynamically.

Score Awarded: 2/3

Q16: What is shallow copy and deep copy?

Your Answer: Detailed correct explanation.

Score Awarded: 3/3

Q17: Explain unary operator overloading.

Your Answer: Unary operators act on a single operand. Examples ++, --, +, -.

Score Awarded: 2/3

Q18: Explain binary operator overloading.

Your Answer: Binary operators are used between two operands.

Score Awarded: 1.5/3

Q19: Member vs friend function operator overloading.

Your Answer: Friend function can be accessed where friend keyword is used while member function is restricted.

Score Awarded: 2/3

Q20: Why is operator overloading useful?

Your Answer: Allows user-defined classes to behave like primitive types.

Score Awarded: 1/1

Q21: Why can't constructors be virtual?

Your Answer: Due to polymorphism.

Score Awarded: 0.5/1

Q22: Why can destructors be virtual?

Your Answer: Enable polymorphic destruction.

Score Awarded: 1/1

Q23: Can a class have multiple destructors?

Your Answer: No, destructor cannot be overloaded.

Score Awarded: 1/1

Q24: Copy constructor vs copy assignment operator.

Your Answer: Copy constructor creates new object; assignment assigns values.

Score Awarded: 0.75/1

Q25: Constructor overloading vs function overloading.

Your Answer: Different parameters/data types.

Score Awarded: 0.75/1

Q26: Purpose of this pointer.

Your Answer: Refers to current object.

Score Awarded: 1/1

Q27: Can constructors/destructors be static?

Your Answer: No, static belongs to class not object lifecycle.

Score Awarded: 1/1

Q28: new/delete vs malloc/free.

Your Answer: new/delete allocate/deallocate, malloc/free are C functions.

Score Awarded: 0.75/1

Q29: Why copy constructor takes reference?

Your Answer: Avoid infinite recursion and stack overflow.

Score Awarded: 1/1

Q30: What is the Rule of Three?

Your Answer: I don't know.

Score Awarded: 0.25/1